GOVIND, 2K21/EE/119

NAS
(EE201)

Network and System

Combination

(i) Active and Passive :-

Active → may not require external power source to operate (Photovoltaic cell, charged inductor) (current source, ~~diode~~ Transistor)

→ [Controls flow of electric current]

Passive → require external power source to drive them. (Resistance, uncharged inductor and capacitor, transformer)

transformer → [power, energy remain constant]

[transistor, → amplify power of a signal] etc

↳ controlling flow of charge

(ii) Unilateral and bilateral ;

Diode + Forward (R = low) -   - Reverse (R = high) +

Unilateral:
- conduction in single direction
- (Diode)
- Polarity exists (Resistance different for both direction)

Bilateral:
- both direction
- Resistance, capacitor, inductor.
- Polarity X

(iii) Linear and Non-linear ;-

Linear → parameters which follow
- superposition
- Homogenity

A system is linear if it hold Principle of Superposition & homogenity

Superposition → (additivity)

Homogenity:

$a x_1(t) \to a y_1(t)$

$b x_2(t) \to b y_2(t)$

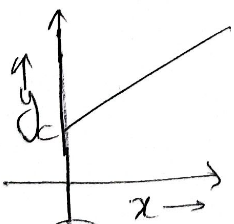
Additivity:

$x_1(t) \to y_1(t)$

$x_2(t) \to y_2(t)$

$x_1(t) + x_2(t) \to y_1(t) + y_2(t)$

$[a x_1(t) + b x_2(t) \to a y_1(t) + b y_2(t)]$



$y_1 = m_1 x + c_1$

$y_2 = m_2 x + c_2$

$y_1 + y_2 = (m_1 + m_2)x + (c_1 + c_2)$

Y   X   C

$[Y = mX + C]$

Difference $[C \neq c_1 + c_2]$ Occurs

Current → not follow vector law of addition

Note: Linear,

$x(t) = 0 \longrightarrow y(t) = 0$

$x[n] = 0 \longrightarrow y[n] = 0$

(iv) Lumped and Distributed :-

Parameter which can't be seperated and Confined at one place.

[V, I → vary x with dimension]

through element

Ex: - Resistance in electrical circuit.

→ change with dimension of element

Ex: Resistance ~~of~~ transmission line.

Note:
"Read after system modelling"

$y(t) = T[x(t)]$ T[ ] denotes

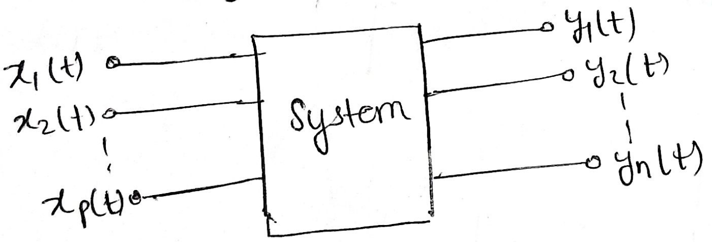
$y[n] = T[x[n]]$ a DT system

T[ ] operates on input signal $x(t)$ to produce output signal $y(t)$.

Systems :

→ combination and interconnection of several components to perform desired network / operation / task

System Modelling : Linear-Time Invariant System



$x_1(t), x_2(t), x_3(t), \ldots x_n(t)$ are input or excitation or cause.

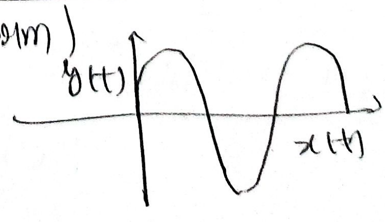
$y_1(t), y_2(t), y_3(t), \ldots, y_n(t)$ O/P or response or effect.

(No break)

Continuous time and discrete time System :-

(Diff. eq^n, Laplace transform)

~~Step input~~ $x(t)$ → Continuous time system → $y(t)$



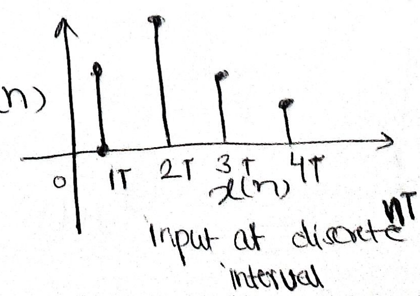
$y(t) = \sin(\omega t)$

I/P & O/P
↳ continuous time function

sampling

$x(n)$ → discrete time system → $y(n)$

(Z transform)
-ation



" "
↳ discrete

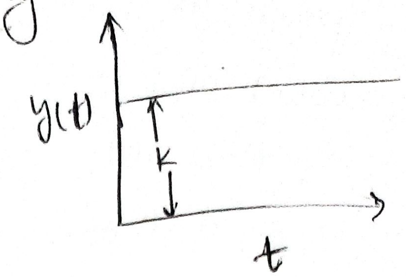
Ex. of continuous f^n :-

1. Step input

Ex: Battery. (D C-source)

$y(t) = 0 \; ; \; t < 0$

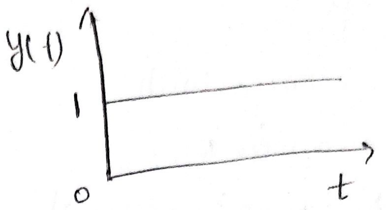
$= k \; ; \; t > 0$



1 (i) unit step

$y(t) = 0 \; ; \; t < 0$

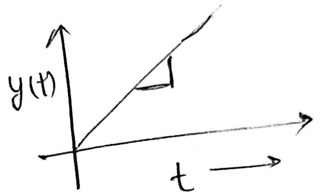
$= 1 \; ; \; t > 0$



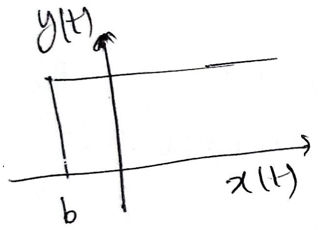
2. Ramp input

$y(t) = kt, \; t > 0$

$0 \; , \; t < 0$

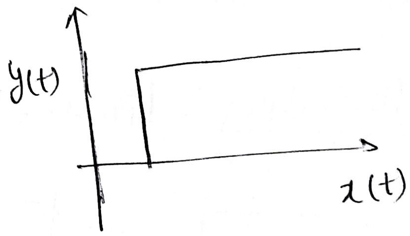


3. Step

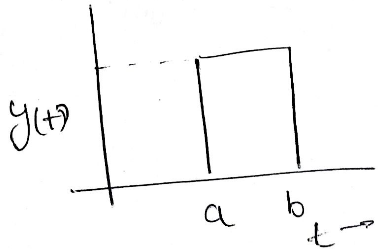


$y = u(t-a) \; , t > a$

$y(t) = u(t+b)$



4. Gauge function :

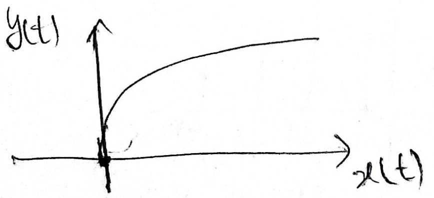


$y(t) = u(t-a) - u(t-b)$

5. Exponential function :-

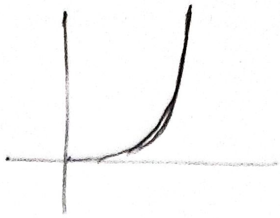
$y(t) = 1 - e^{-t}$

$= 0 \quad , t < 0$



6. Parabolic f^n :-

$y = t^2$



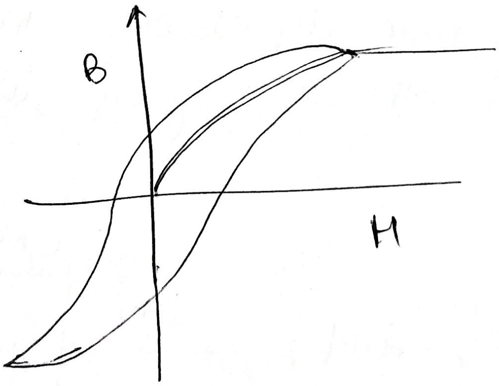
Time invariant and time varying system:

A system is a time invariant or fixed if it's input, output relation-ship doesnot change with time otherwise it is said to be time variant system.

[Note: $y(t) = T[x(t)]$
for d if,
$y(t-d) = T[x(t-d)]$
→ time invariant
→ cont. $d \in R$
disc. $d \in Z^{+}$

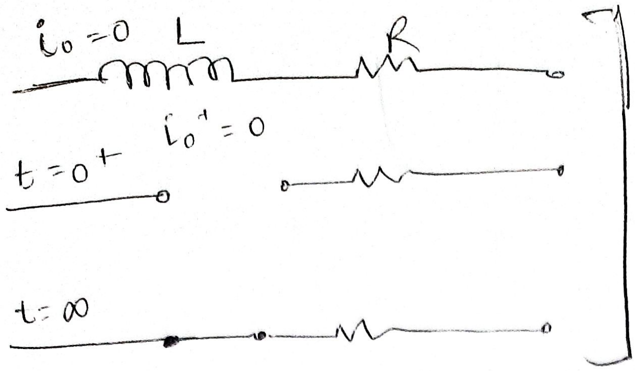
Instantaneous and Dynamic system:-

⇒ A system for which output is a function of input at present time only is said to be instantaneous system or memory-less or zero memory system.

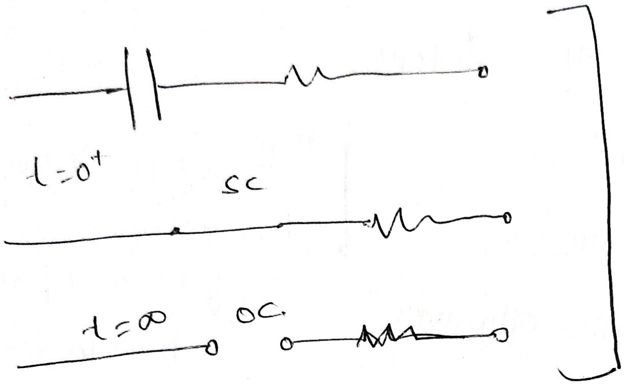


for all t, there exist a function, f_t such that
$y(t) = f_t(x(t))$
where,
$y(t) = T[x(t)]$
→ memoryless.
fn → disc.

⇒ On the other hand dynamic system is one whose output depends on past or future values of input or past output in addition to present input.



$t = 0 \leftrightarrow \infty$
middle time
↓
transient state



for all $t \leq T$,
$\Rightarrow x_1(t) = x_2(t)$
$\Rightarrow y_1(T) = y_2(T)$
↳ Causal
$y_1(t) = T[x_1(t)]$
$y_2(t) = T[x_2(t)]$

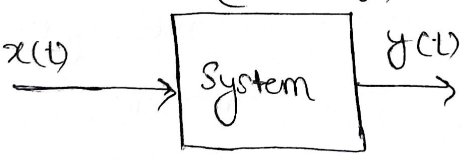
(non-anticipative)

Causal and Non-causal System :

- If a system is said to be causal if output of system at any time depends on value of input at present and in the past i.e output doesn't anticipate future value.
(t=0)

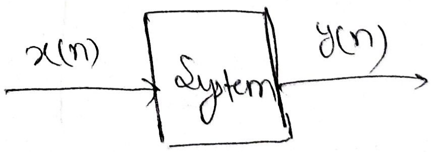
- Output of system at any time depends only input at present and past value of input otherwise system is said to be non-causal

(Causal)



and

(Non Causal)



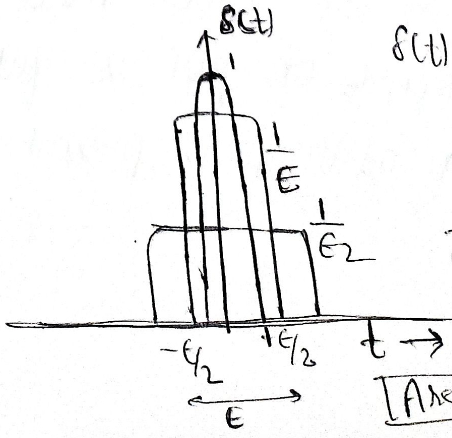
(Continuous)

$y(t) = x(t+1)$

non causual
if it's output depends on future value of input

$y[n] = x[n]$

(discrete)



$\delta(t) = \begin{cases} 1, & t = 0 \\ 0, & t \neq 0 \end{cases}$

$\frac{1}{E} \times E = 1$

[Area = 1]

Impulse Response :- $h(t)$ of a continuous-time LTI system (T)

$x(t)$

$$h(t) = T\{\delta(t)\}$$

I/P $x(t)$ is represent to an arbitrary input

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau$$

$y(t) \to$ O/P , $\quad y(t) = T\{x(t)\}$

$$y(t) = \int_{-\infty}^{\infty} x(t)\, T\delta(t-\tau)\,d\tau$$

$\because$ System is Linear Time Invariant

$$h(t-\tau) = T\delta(t-\tau)$$

$$\boxed{y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\,d\tau}$$

~~Convultion~~ Convolution integral,

$$\boxed{y(t) = x(t) * h(t)}$$

$$\boxed{y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau}$$

Property of convolution integral :-

1. Commutative

$$x(t) * h(t) = h(t) * x(t)$$

2. Associative

$$\{x(t) * h_1(t)\} h_2(t) = x(t) *$$

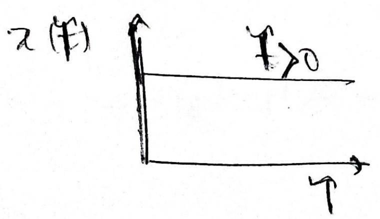
3. Distributive

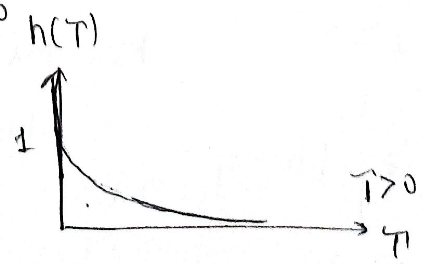
$$x(t) * \{h_1(t) + h_2(t)\} = x(t) * h_1(t) + x(t) * h_2(t)$$

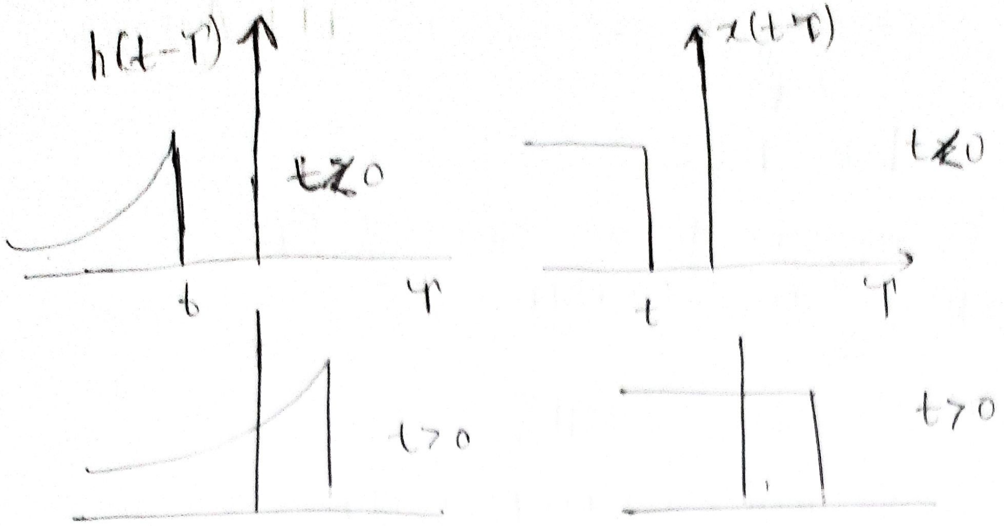
Ex: The input $x(t)$ and impulse response $h(t)$ of a continuous LTI system is given by, $x(t) = u(t)$, $h(t) = e^{-\alpha t}$, $\alpha > 0$, then evaluate output.

Sol^n

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\,d\tau$$







Function $h(T)$ and $h(t-T)$ are shown for $t<0$ and $t>0$, we see that for $t<0$, $h(T)$ and $h(t-T)$ ~~hence~~ donot overlap each other. Hence it will never overlap ~~fo~~. for $t>0$ they overlap from $T=0$ to $T=t$, hence $y(t)$ for $t>0$ will be

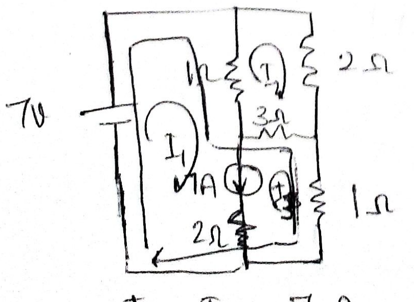
$$y(t) = \int_0^t e^{-\alpha(t-T)}\,dT$$

$$y(t) = \int_0^t \frac{e^{\alpha T}}{e^{\alpha t}}\,dT$$

$$= \frac{1}{\alpha e^{\alpha t}}\left(e^{\alpha t}-1\right) = \frac{1}{\alpha}\left(1-e^{-\alpha t}\right)$$

Supermesh: ~~When in a mesh, branch comprises~~ Concept
A path around a supermesh doesnot pass through current source. Path around each mesh contained within a supermesh passes through a current source. Total no. of eqn's required for a supermesh is equal to no. of meshes contained in supermesh. Supermesh requires 2 mesh eqn.

KVL for outer loop



$$7 - 1\cdot(I_1 - I_2) - 3(I_3 - I_2) - 1\cdot I_3 = 0$$

KVL for mesh(2):-

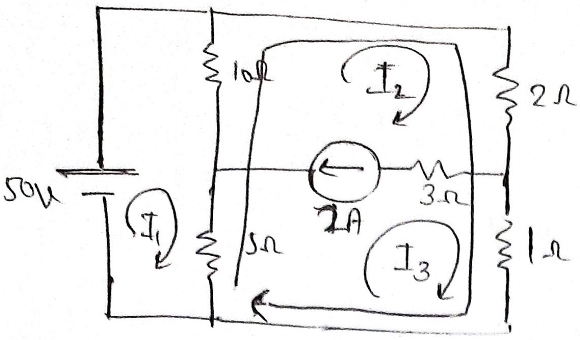
$$-1(I_2 - I_1) - 2I_2 - 3(I_2 - I_3) = 0$$

$I_1 - I_3 = 7\text{ A}$

$I_1 = 9\text{ A}$

$I_2 = 2.5\text{ A}$

$I_3 = 2\text{ A}$



Find current in 5Ω.

Sol^n

$I_2 - I_3 = 2 \longrightarrow (1)$

KVL in outer loop:-

$10(I_2 - I_1) + 2I_2 + I_3 + 5(I_3 - I_1) = 0 \longrightarrow (2)$

KVL in mesh(1):-

$10(I_1 - I_2) + 5(I_1 - I_3) = 50 \longrightarrow (3)$

$I_1 = 20A$

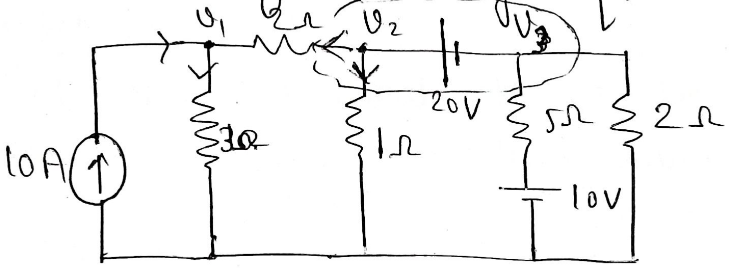
$I_2 = 17.33A$

$I_3 = \cancel{2A}\ 15.3A$

Midsem: Unit – 1, 3

Supernode concept: Nodes that are connected to each other by a voltage source but not to the reference node by a path of voltage source from a supernode. Supernode requires voltage eq^n and remaining node voltage eq^n.

Determine current in 5Ω.



Sol^n $V_2 - V_3 = 20V. \longrightarrow (1)$

Apply KCL at supernode:-

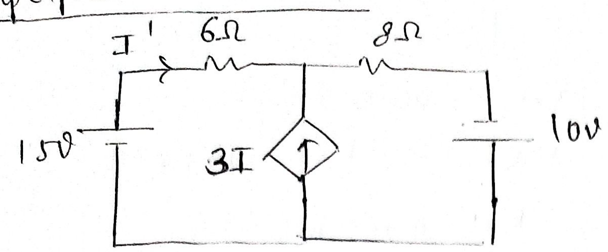
$$\frac{V_2 - V_1}{2} + \frac{V_2}{1} + \frac{V_3 - 10}{5} + \frac{V_3}{2} = 0 \longrightarrow (2)$$

KCL at node 1:-

$$10 + \frac{V_2 - V_1}{2} = \frac{V_1}{3} \longrightarrow (3)$$

NOTE: "Applicable on voltage source having no. resistance.

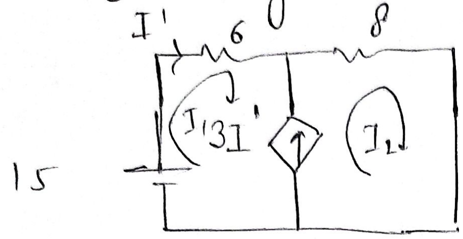
Superposition theorem :-



Find current in 6Ω.

$I = 0.65A$

Solⁿ Considering 15V source only



$I_2 - I_1 = 3I'$

$I_2 - I_1 = 3I_1$

$I_2 = 4I_1$ — ①

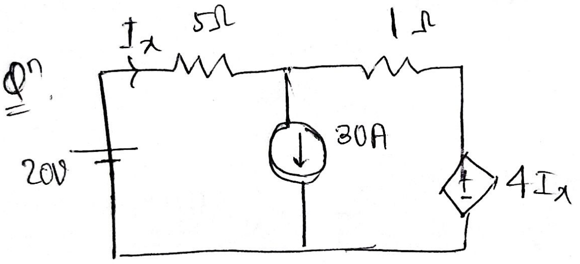
$I_1 = I'$

Apply KVL, $15 - 6I_1 - 8I_2 = 0$

$6I_1 + 8I_2 = 15$ — ②

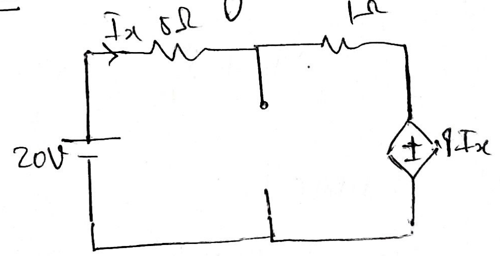
Solving ① & ② :-

Qⁿ.



Find I_x

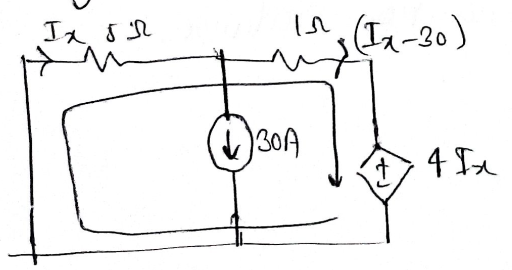
Solⁿ Considering 20V source only.



$6I_x + 4I_x - 20 = 0$

$\boxed{I_x' = 2A}$

Considering 30A source only,



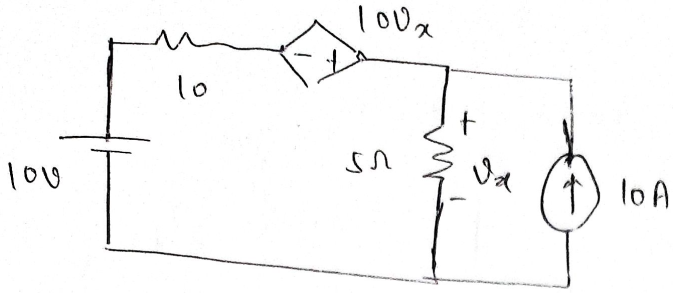
$(I_x - 30) \times 1 + 4I_x + 5I_x = 0$

$10I_x = 30$

$\boxed{I_x'' = 3A}$

$I_x = 5A$

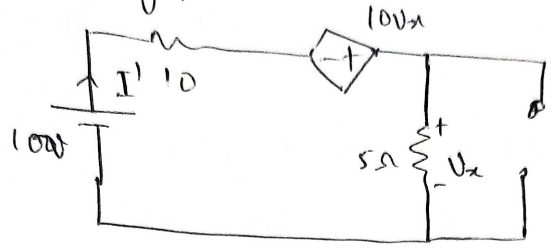
Q.



Find current in 10Ω

Sol^n

Considering 10V source.



$I' = \frac{100 - 15I' + 100V_x}{15}$

$V_x = 5 \times I'$

$I' = \frac{100 - 10I' + 5I'}{15}$

$-20I' = 100$

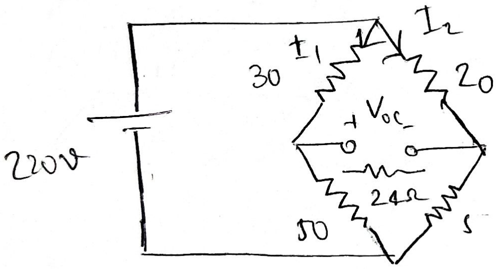
$\boxed{I' = -5A}$

$15I' = 10 - 55I'$

$\boxed{I' = \frac{10}{7} A}$

Thevenin's theorem :-

⇒ It stated that any 2 terminal ckt can be replaced by equivalent resistance and equivalent voltage source.



Find current in 24Ω.

Sol^n

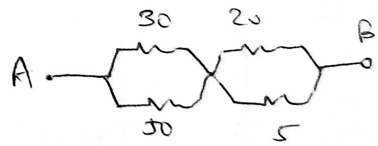
$I_1 = 2.75 A$

$I_2 = 8.8$ Amp.

$\boxed{V_{th} = \cancel{264V}\ 93.5V}$

$R_{eq} =$

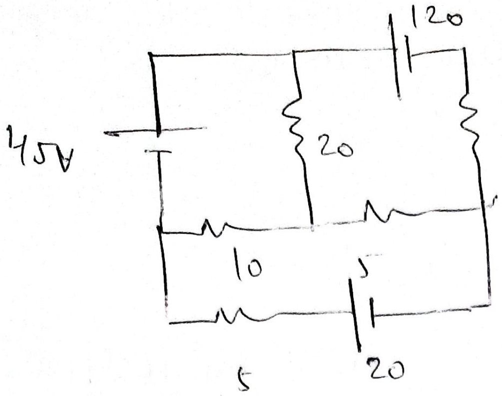
$R_{th} = 22.75 \Omega$



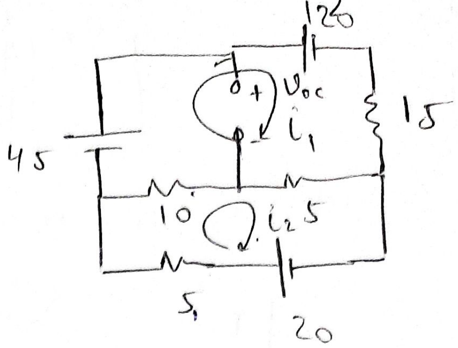
$R_{th} = 22.75 \Omega$

$R_{eq} = \frac{80 \times 25}{80 + 25} =$

Current in 20Ω = ?



Sol^n



$V_{th} = 63V$

$R_{th} = 4.67\Omega$

$30i_1 - 15i_2 - 45 + 120 = 0$

$30i_1 - 15i_2 = 75$ —— (1)

$\cancel{5}i_1 - 3i_2 = 15$ — (i) ~~×3~~

$20i_2 - 15i_1 = 20$ —— (2)

$4i_2 - 3i_1 = 4$ — 2(i) ×2

$5i_2 = -23$

$i_2 = 4.6A$

$i_1 = \frac{4 \times 3.612}{3} = 9.2A$

~~$11i_2 = 45 + 20$~~

~~$i_2 = \frac{65}{11} A$~~

$i_1 = \frac{4(i_2-1)}{3}$

$= \frac{4}{3} \times \frac{54}{11}$ 18

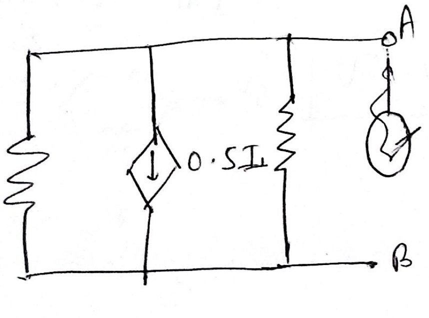
$\boxed{i_1 = \frac{72}{11} A}$

~~V_{th} 120 − 15 × $\frac{72}{11}$ − $\frac{3}{11}$ × 5 = 0~~

~~V_{th} = 120 + 98.18 + 3.18~~

$V_{th} - 120 - 15 \times 9.8$ -

Q^4



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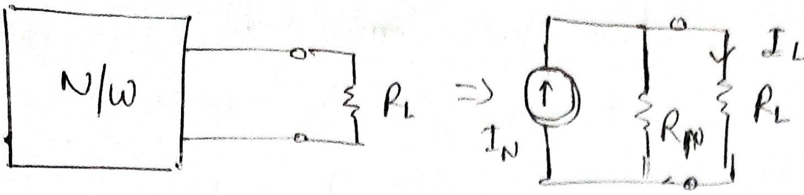
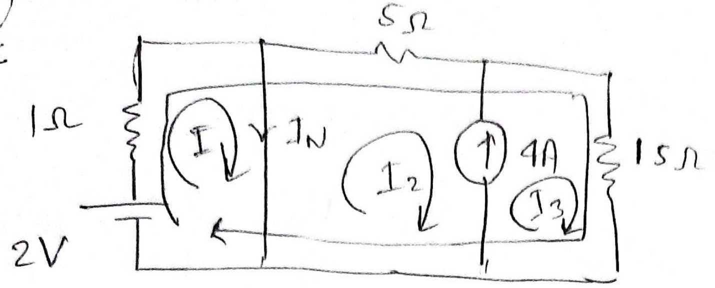
Norton:



$$\boxed{I_L = \frac{I_N \times R_N}{R_N + R_L}}$$

Q^n



$I_1 - I_2 = I_N$

$I_3 - I_2 = 4$ — (2)

Loop (1) :-

$2 - I_1 \times 1 = 0$

$\boxed{I_1 = 2A}$

$2 - 1.I_1 - 5I_2 - 15I_3 = 0$

$2 \cdot 2 - 5I_3 - 15I_3 = 0$

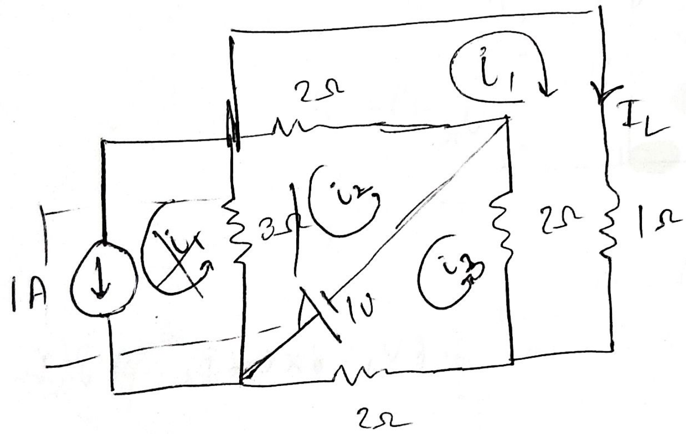
$I_2 = -3I_3$

$4I_3 = 4$

$I_3 = 1A$

$\boxed{I_2 = 3A}$

Q^n



Find current in 1Ω

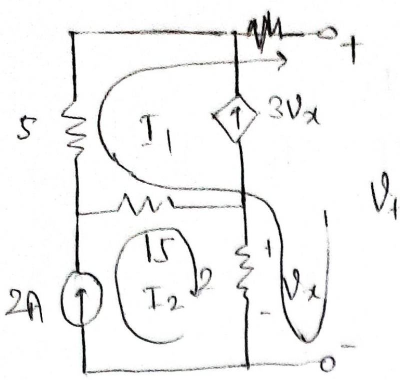
$i_1 = 1A$

$i_2 =$

Sol^n

$R_N = \frac{11}{5}\,\Omega$

Q^n



Find norton equivalent across AB-terminals.
I_N & R_N to be determined or
also V_{th} or R_{th}

$\rightarrow I_1 = 1.69A$

$V_{th} - 2I_2 + 15(I_1 - I_2) + 5I_1 = 0$

$R_N = \frac{V_{th}}{I_N} = \frac{274}{1.69} = 162.13$

$V_{th} - 4$

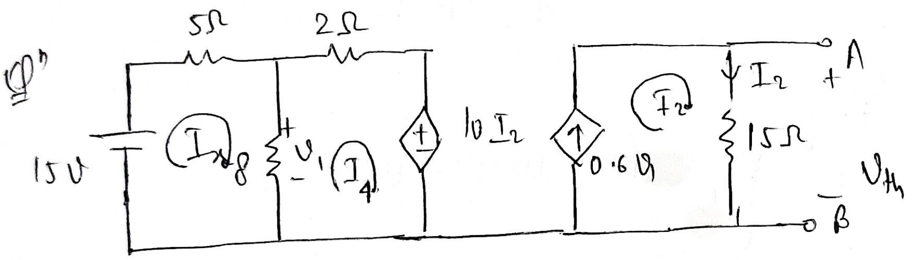
$I_1 = -3V_x$

$V_x = 2I_2$

$V_{th} = 274V$

$\boxed{I_2 = 2A}$

Q^n



① $15 - 5I_x - 8(I_x - I_4) = 0$

$V_1 = 8(I_x - I_4)$

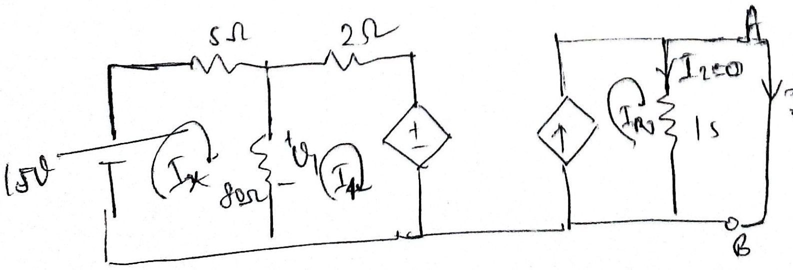
$0.6V_1 = \cancel{8 \times 0.6 I_x} = 4.8(I_x - I_4)$

② $-8(I_4 - I_x) - 2I_4 - 10I_2 = 0$

③ $V_{th} = 15I_2$

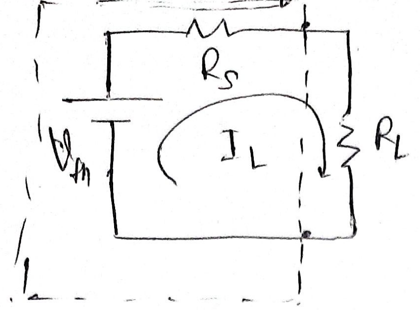
$I_2 = 0.6V_1$

$V_{th} = 15 \times 0.6V_1 = 9V_1$



$\boxed{I_N = 2.16A}$

Maximum Power Transfer Theorem :-



$$I_L = \frac{V_{th}}{R_S + R_L}$$

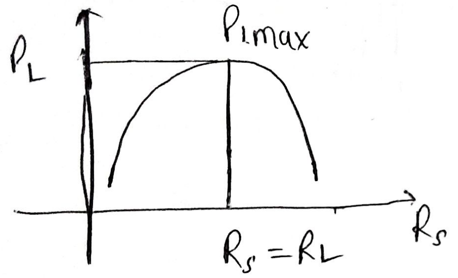
$$P_L = I_L^2 \cdot R_L$$

$$= \frac{V_{th}^2}{(R_S+R_L)^2} \times R_L$$

P_L will be max. if $\frac{dP_L}{dR_L} = 0$ &

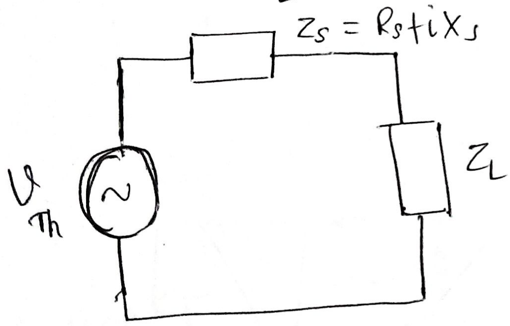
$$\frac{d^2P_L}{dR_L^2} = -ve$$

$$\frac{dP_L}{dR_L} = V_{th}^2 \left[\frac{(R_S+R_L)^2 * 1 - R_L \cdot 2(R_S+R_L)}{(R_S+R_L)^4}\right] = 0$$



$$(R_S+R_L)^2 = 2R_L(R_S+R_L)$$

$$\boxed{R_S = R_L}$$



$$Z_S = Z_L^*$$

$$Z_S = R_L - jX_2$$

$$\boxed{jX_S = -jX_L}$$

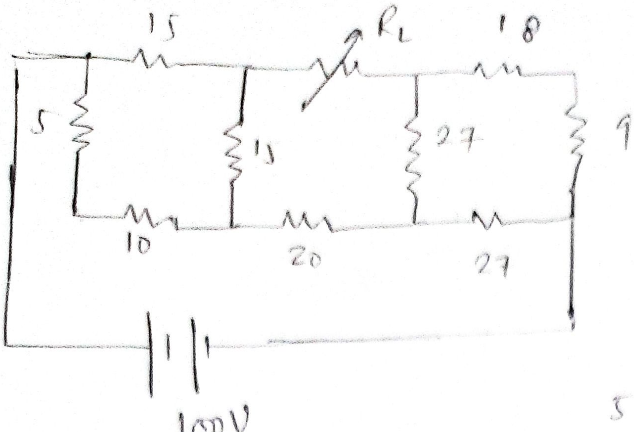
$$P_{Lmax} = \frac{V_{th}^2}{4R_L^2} \times R_L = \frac{V_{th}^2}{4R_L}$$

$$O/P \text{ Power} = \frac{V^2}{4R_L}$$

$$\text{Input power} = \frac{V^2}{4R_L} + \frac{V^2}{4R_L} = \frac{V^2}{2R_L}$$

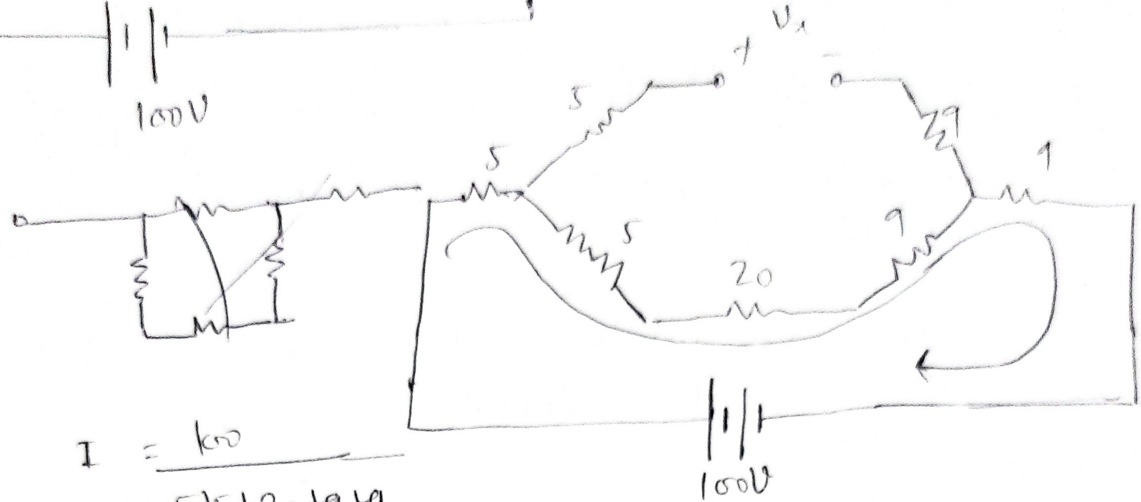
$$\eta\% = \text{efficiency}\% = \frac{O/P}{I/P} \times 100 = \underline{\underline{50\%}}$$

Q^n



Sol^n

$\Delta \to Y \to \frac{R}{3}$

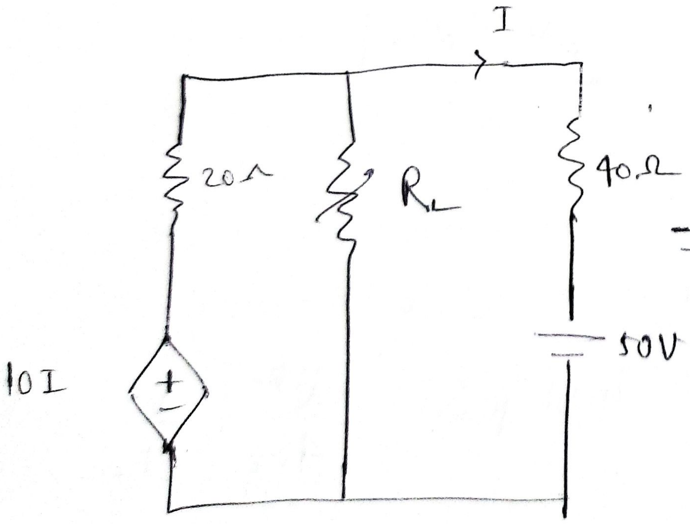


$$I = \frac{100}{5+5+20+9+9}$$

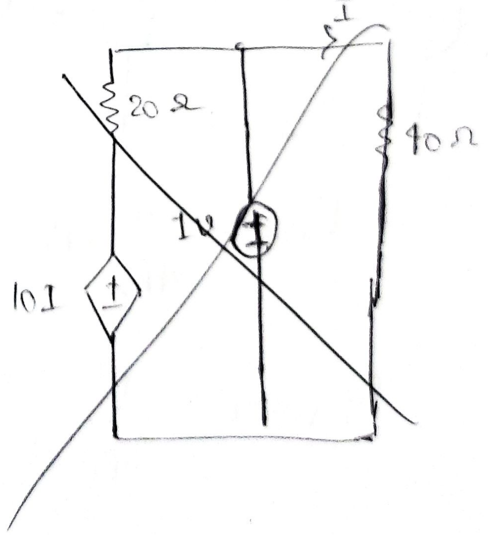
$$= 2.07A$$

$$V_{th} = (5+20+9) \times 2.07$$

R_{th} =



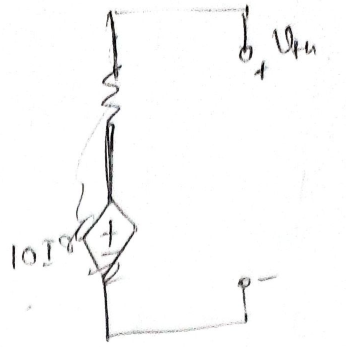
⇒

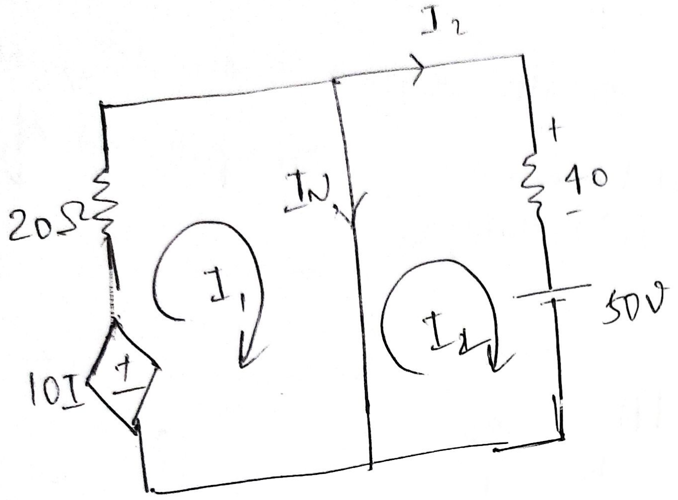


$60I + 50 - 10I = 0 \Rightarrow \boxed{I = -1A}$

$V_{th} - 40(-1) - 50 = 0$

$\boxed{V_{th} = 10V}$





$I_1 - I_2 = I_N$ — (1)

$I = I_2$

$10I - 20I_1 = 0$

$10I_2 = 20I_1$

$I_2 = 2I_1$

~~$-1.25 + 0.625 =$~~

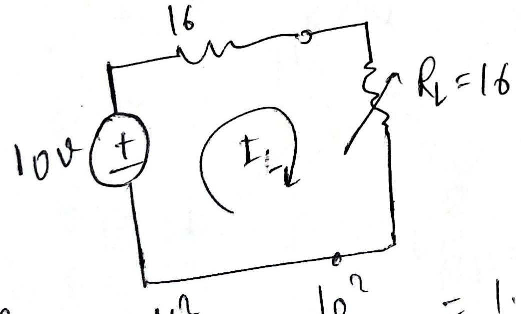
$50 = -40I_2$

$I_2 = -1.25$

$I_1 = \frac{I_2}{2} = -0.625A$

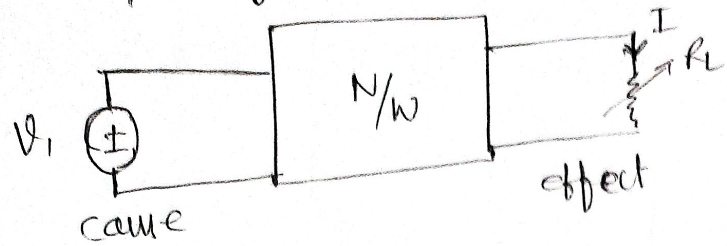
Note :- Apply 1A or 1V when there are only dependent sources.

$R_N = \frac{V_{th}}{I_N} = \frac{10}{0.625} = 16A$



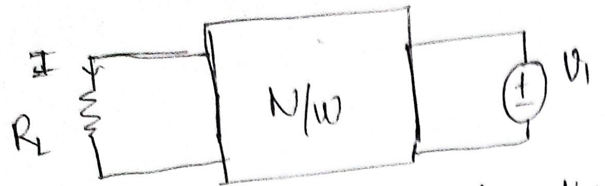
$P_{Lmax} = \frac{V^2}{4R_L} = \frac{10^2}{4\times16} = 1.56W$

Reciprocity theorem :-

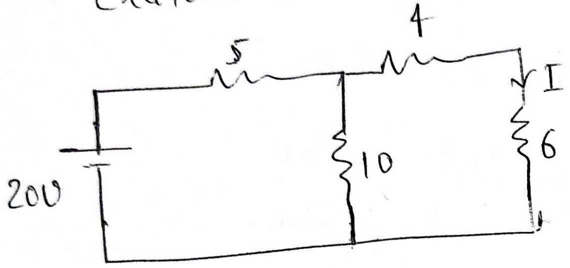


$$\frac{V_1}{I} = \frac{V_1}{I} = K$$

$$\frac{\text{cause}}{\text{effect}} = K$$

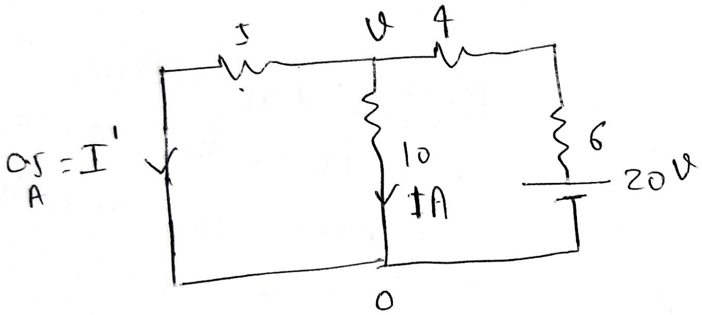


⇒ In a linear bilateral active single source network, ratio of excitation to response remain same when position of excitation and response are interchanged.



$I = 1A$

$K = \frac{20}{1} =$

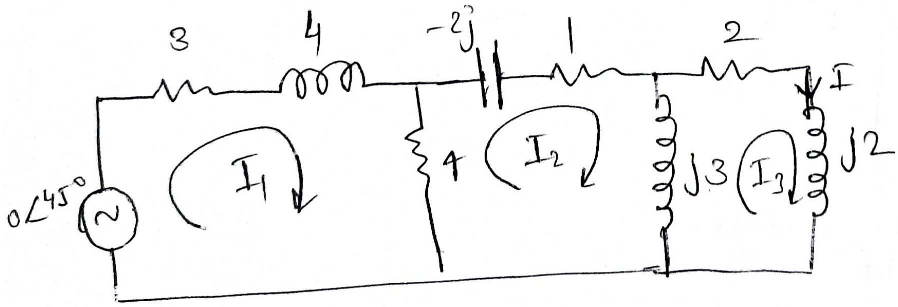


$R_{eq} = (10 \| 5) + 10$

$= \frac{10 \times 5}{15} + 10 = \frac{10}{3} + 10$

$= \frac{10 \times 4}{3}$

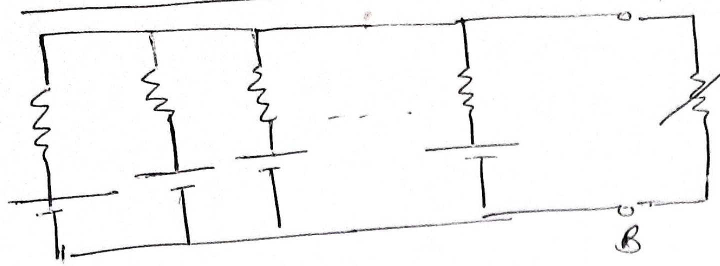
$I = \frac{20}{\frac{10 \times 4}{3}} = \frac{3}{2} A$



$I_3 = I$

$I_3 = 0.704 \angle 30.72^\circ$

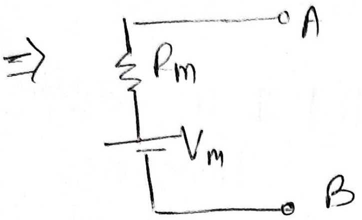
Milliman's theorem :—



$$V_m = \frac{V_1G_1 + V_2G_2 + \dots + V_nG_n}{G_1 + G_2 + G_3 + \dots + G_n}$$

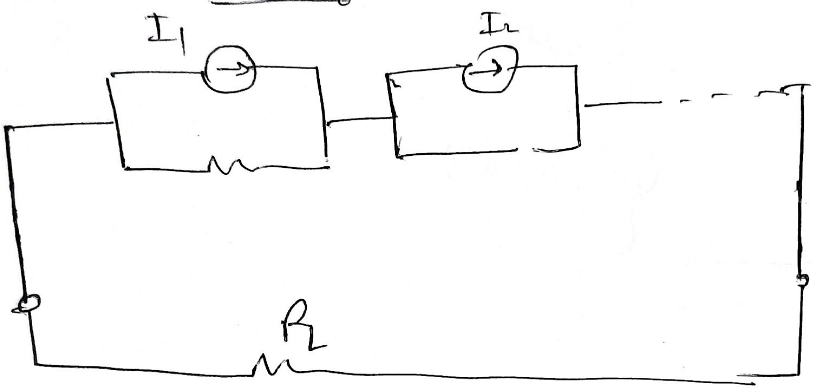
$$R_m = \frac{1}{G_m} = \frac{1}{G_1 + G_1 + G_2 + G_4 + \dots + G_n}$$

⇒

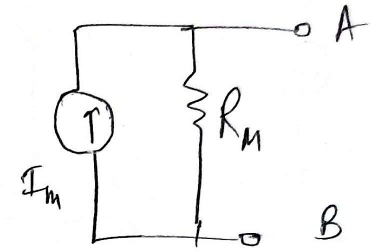


$Z = R + jX$

$Y = G + jB$ — Susceptance

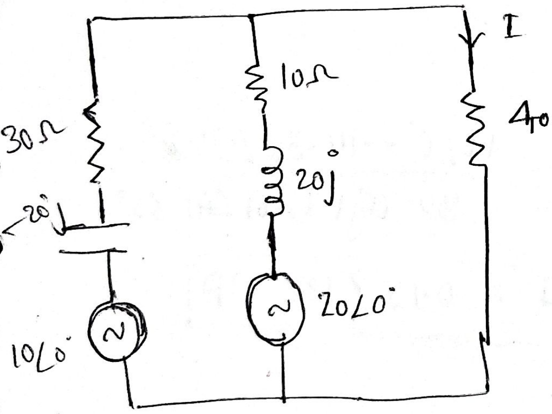


≡



$$I_m = \frac{I_1Z_1 + I_2Z_2 + \dots + I_nZ_n}{Z_1 + Z_2 + \dots + Z_n}$$

$$R_m = Z_1 + Z_2 + Z_3 + \dots + Z_n$$



Find I use millman's theorem.

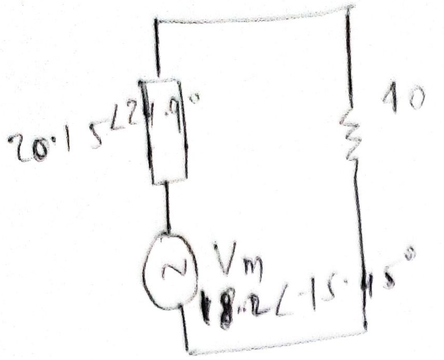
$$V_m = \frac{10\angle 0^\circ \times \frac{1}{30 - 20j} + 20\angle 0^\circ \cdot \frac{1}{10 + 20j}}{\frac{1}{30 - j20} + \frac{1}{10 + 20j}}$$

$V_m = 18.2\angle -15.95^\circ$

$Y_m = \frac{1}{Z_m} =$

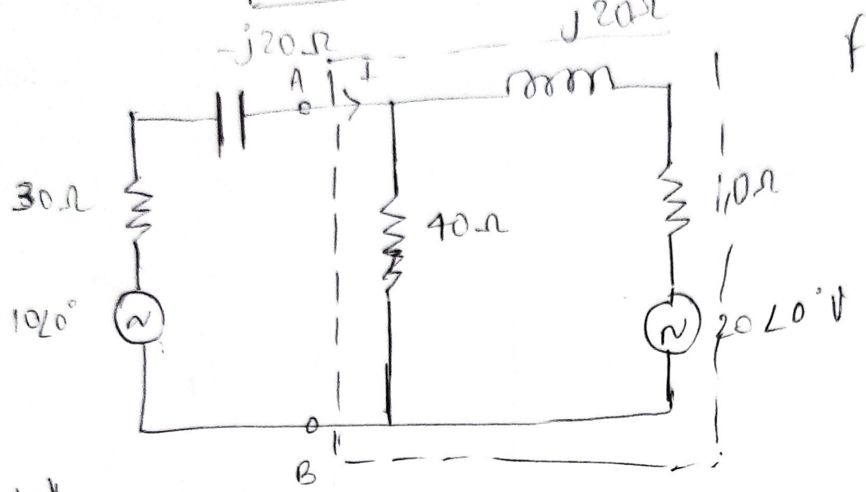
$Y_m = Y_1 + Y_2$

$Z_m = \frac{1}{Y_1 + Y_2} = \frac{1}{\ } = 20.15\angle 29.74^\circ$



$\boxed{I_L = 0.31\angle -25.81^\circ\ A}$

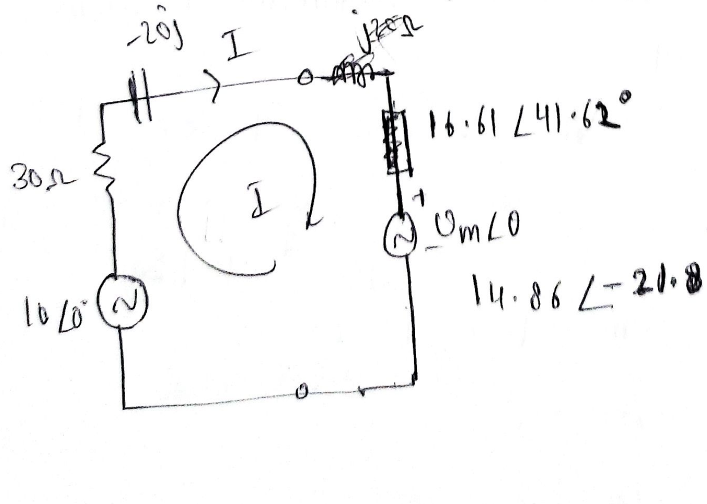
find I use millman's theorem.



Sol^n

$$V_{AB} = \frac{\frac{20\angle 0^\circ}{10+20j} + 0}{\frac{1}{10+20j} + \frac{1}{40}} \quad = \cancel{10\angle 0^\circ - (30-20j)I}$$

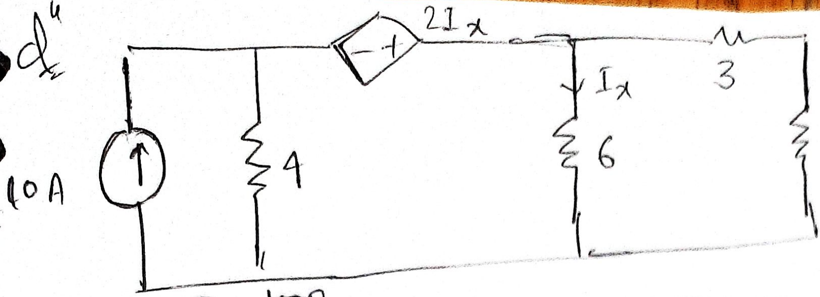
$$Z_m = \frac{1}{\frac{1}{10+20j} + \frac{1}{40}}$$



$$I = \frac{10\angle 0^\circ - 14.86\angle -21.8}{30-20j + 16.61\angle 41.63^\circ}$$

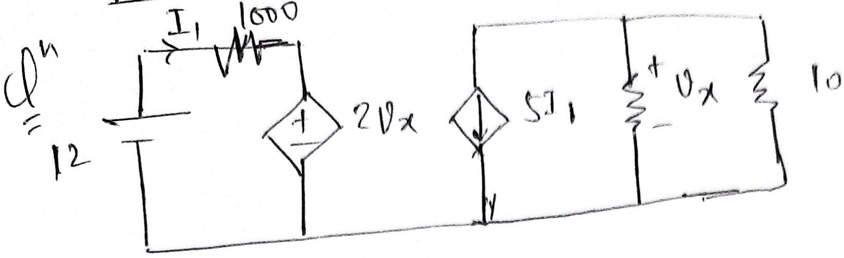
$\boxed{I = 0.15\angle 136.5^\circ A}$

Q"



Find R_L for max. power R_L and also find max. power.

Q"



Find current through 10Ω.